Pull the Line -- Instructor Notes

Leverage, influence, and the points that move a regression line

Overview

Draft, class number not yet assigned. Runs after least-squares regression has been taught. Students play a browser game and record every commitment and result on the paper handout (`PullTheLine.pdf`, 5 pages). Part 5 is a skeleton notebook (`PullTheLine_Skeleton.ipynb`, with `data/cricket_thermometer.csv` and `data/plant_dataset_1.csv` to `_3.csv`) that reproduces the game's numbers and runs the probe on every point.

Game: `https://jonathan-nyquist.github.io/EDS_Artifacts/pull-the-line/`. Adding `#r2` or `#r3` to the URL opens that round directly, so any round can be skipped. Reloading the page clears a student's answers; there is no data collection.

Data: Rounds 1–2 use the 15-point cricket dataset from Classes 18–19 (`cricket_thermometer.csv`, points numbered in file order). Round 3 uses three synthetic datasets (seed 2026, the first three draws, not hand-picked), built by `verify_round3b_shift20.py`. Every number below was produced by the scripts in `LeverageGame/` run in `ds313`, and the game reproduces them.

Arc: fit by eye and compare with least squares – commit to the one point that will tilt the line most, discover it is the point farthest from mean x (leverage) – hunt a planted point using leave-one-out probes, discover that a far-out bad point hides in the residual plot and that a far-out real point can move the line without being wrong (influence is not error) – close on when deleting a point is justified.

Learning objectives

Students should be able to:

1. Explain that a point's leverage depends only on how far its x-value is from mean x.
2. Explain why an influential point can have a small residual: it pulls the line toward itself.
3. Distinguish leverage (potential), influence (actual slope change), and being off the trend (miss from the line fit without the point), using leave-one-out refits.
4. Argue that "influential" is not by itself a reason to delete a data point.

Not taught: the leverage formula h_i, Cook's distance, studentized residuals, or any formal outlier test. The "miss from the line fit without it" is a leave-one-out (deleted) residual, but the handout never uses that term. Don't introduce it unless a student asks.

Timing

Part	Min	Content
Setup	3	Open the game, teams of two, one handout per team
1	6	Fit by eye – commit, lock, SD-line discussion
2	7	Tilt the line – commit to a point, reveal, explore
3	12	Find the plant – three datasets with two probes each, then Q3.1–Q3.3
4	7	Would you delete it? – discussion
5	35	Notebook: move-one-point loop, leave-one-out loop, all three game datasets

That totals 70 minutes. The split reflects how this class works: point-and-click parts go fast, coding is where students bog down. The time is weighted toward Part 5, and its three loops are the likeliest place to run over. If time runs short, cut Part 5C first. 5A and 5B carry the method. If a team is stuck on the 5B loop, point them to

`leave_one_out(t)` in 5C, which contains the same loop, and have them run it on `crickets`. If Part 4′s discussion works better as the last word, run it after Part 5.

Part 1 – Fit by eye

Least-squares slope / intercept	3.291 °F per chirp/sec, 25.23 °F
Correlation r	0.835
SD line slope (SD_{temp} / SD_{chirps})	3.941
Least-squares SSE (the score's denominator)	190.55

Score = student's SSE / 190.55. Only 1.00 exactly is perfect, and nothing can score lower. The handles start flat at mean temperature, so a student who never touches them scores badly.

Caution: Q1.1 asks for a steeper/shallower prediction. It is tempting to tell students that people systematically fit lines too steep, closer to the SD line (Mosteller et al., 1981). That claim has **not** been verified for these notes. Present Q1.3 as a question about the class's own lines, not as a known result, until the citation is checked.

Q1.3 key. Least squares slope = r × SD-line slope: 0.835 × 3.941 = 3.291. The SD line runs along the long axis of the cloud and treats chirps and temperature symmetrically. Least squares answers a one-way question – predict temperature from chirps – and because $|r| < 1$ its predictions are pulled toward the mean temperature. A student whose eye-fitted line lands near 3.94 has drawn the trend of the cloud, not the best prediction line. That is the regression effect, and worth naming if the class's lines are visibly steep. Take a quick show of hands: "closer to 3.29 or to 3.94?"

Part 2 – Tilt the line

Caution: The game's reveal names the best point. Check that each team has written its Q2.1 choice and reason before anyone clicks **Move my point**.

Moving point i up by 10 °F changes the slope by exactly $10(x_i - \overline{x})/(n \cdot \mathrm{SD}_x^2)$, with $\overline{x} = 16.653$ and $n \cdot \mathrm{SD}_x^2 = 40.54$. This was checked against a direct refit for all 15 points, with agreement to 10^{-15}. Moving a point up steepens the line if it is right of mean x and flattens it if it is left. The residual never enters the formula.

Rank	Point	x	Slope change	Rank	Point	x	Slope change
1	1	20.0	+0.825	9	2	16.0	−0.161
2	3	19.8	+0.776	9	13	16.0	−0.161
3	15	14.4	−0.556	11	12	17.2	+0.135
4	7	14.7	−0.482	12	10	16.2	−0.112
5	4	18.4	+0.431	13	5	17.1	+0.110
6	11	15.0	−0.408	13	8	17.1	+0.110
7	9	15.4	−0.309	15	14	17.0	+0.085
8	6	15.5	−0.284				

Ties (points 2 and 13, points 5 and 8) share a rank in the game.

Q2.3 key. Point 9 has the largest residual (−6.52 °F) but ranks only 7th. The top four points are the four farthest from 16.65 chirps/sec, two on each side. Watch for teams choosing point 9 (biggest residual) or point 3 (highest temperature) – a guess, not yet observed. Ask anyone who chose point 9 why. Their answer is the misconception this round exists to break.

Note: The leverage note at the end of Part 2 is a definition, not a discovery. Have students read it aloud **after** Q2.3, not before.

Part 3 – Find the plant

Each dataset has 13 ordinary points, one point near mean x, and one isolated point at 21–22 chirps/sec. Either the far point (Datasets 1 and 3) or the near-center point (Dataset 2) was moved 20 °F. Because there is always a far point, “always accuse the far one” does not work.

Dataset	Plant (x)	Moved	Plant’s residual rank	Slope before	Slope after
1	21.5 (far)	up	2nd	3.804	6.170
2	16.7 (near)	down	1st	4.339	4.635
3	21.2 (far)	up	2nd	3.519	5.558

Probe values students should copy (slope change, miss without it):

Dataset	Point	Slope change	Miss without it (°F)
1	x = 21.5, the plant	+1.93	+16.3
1	x = 18.3, the largest residual	−0.26	−8.0
2	x = 16.7, the plant	+0.32	−21.4
2	x = 21.5, real data	−0.96	−7.3
3	x = 21.2, the plant	+1.25	+12.2
3	x = 16.5, the largest residual	−0.12	+7.8

The intended strategy is to probe the far point and the largest-residual point, then accuse whichever misses its refit line by more. Across 2,000 simulated datasets of each kind:

Approach	Far plant	Near plant
Accuse the largest residual	41.7%	100%
Two-probe strategy, 20 °F shift (used)	97.8%	99.5%
Two-probe strategy, 15 °F shift	88.4%	94.4%

The 20 °F shift was chosen so the evidence nearly always resolves. The cost is that the plant is often visible by eye in the scatterplot. That is why the handout requires the probe table: grade the evidence and the reasoning, not just “Correct? Yes.”

Caution: The method is not infallible. For an innocent far point, the miss without it has a median of 3.9 °F, but its 95th percentile is 11.5 °F. A point far from the others is predicted poorly when left out, because the refit line is extrapolating. If a student asks why the far point’s miss is “large-ish” even when it is innocent, that is the reason, and it is a good question.

Q3.1 key. In Dataset 2, the real far point changed the slope three times as much as the plant (−0.96 vs +0.32), but missed the line fit without it by far less (7.3 vs 21.4 °F). Moving the line a lot shows leverage plus some influence. It is not evidence of an error.

Q3.2 key.

	Slope change	Miss without it
Moved point, far from center	big	big
Moved point, near center	small	big
Real point, far from center	can be big (Dataset 2: −0.96)	small

Accept “small or medium” for the third row’s slope change. The point is that it can be big.

Q3.3 key. A residual is measured from a line the point helped fit. A far-out point has enough leverage to drag the line toward itself, shrinking its own residual: 6.8 °F for the Dataset 1 plant, against 16.3 °F once it is removed. Removing the point lets the other 14 set the line, so its distance from that line is an honest measure of how off-trend it is.

> **Note:** Callbacks, if time allows. Class 17′s Cars 2 had the leverage while Onward had the bigger residual (−18.0 vs +10.8). This round is the diagnostic that Class 17 promised. Class 16′s 2-SD rule failed for the same structural reason: the reference statistics included the contamination they were meant to detect.

Part 4 – Would you delete it?

Q4.1 key (discussion). A strong answer:

- checks the sample’s record (field notes, lab QA, units, transcription);
- asks whether the site differs in kind from the others (a new input, different land use) – if it does, it is a different population, not an error;
- refits with and without the point and reports both.

A good reason to remove a point is a documented error, or the point falling outside the scope of the question being asked. “It is influential” or “it changes my slope” is not a reason. Removing it for that reason is choosing the answer after seeing the data, which ties back to fixing analysis choices before looking at results.

> **Caution:** Q4.1 is a hypothetical. Don’t present it as a real Pennypack sample or attach real numbers to it unless they are pulled from the course data and verified first.

Part 5 – Leave one out (notebook)

Executed end to end in `ds313`. The Class 19 copy of `cricket_thermometer.csv` (rounded values, `chirps_per_sec` / `temp_f` headers) reproduces every game number to the displayed precision. The toolbox functions are copied from Class 19 (`slope_of(t)`, `intercept_of(t)` on a table). Student blanks: the 3-row test loop, the 15-row loop, the leave-one-out loop, and Datasets 2–3.

Q5.1 key. +0.825, matching the game.

Q5.2 key. A perfectly straight line through zero at mean x = 16.65 chirps/sec, slope $10/(n \cdot \mathrm{SD}_x^2) = 0.247$ per chirp/sec. Points right of the mean steepen the line, points left of it flatten it, and the farther out, the bigger the effect. That is leverage.

Q5.3 key. No clear pattern. Point 9 (residual −6.52) has slope change −0.309, smaller than several points with small residuals.

Q5.4 key.

Point	Residual	Miss without it	Miss ÷ residual
1	−2.45	−3.73	1.52
3	+2.90	+4.21	1.45
9	−6.52	−7.28	1.12

Points 1 and 3 are the two farthest from mean x, so they had pulled the line hardest toward themselves. Point 9 is near the middle and barely changes.

Note: Not for students unless asked: miss ÷ residual is exactly $1/(1-h_i)$, where $h_i = 1/n + (x_i - \overline{x})^2/(n \cdot \mathrm{SD}_x^2)$ is the leverage. For point 1, $h = 0.067 + 0.276 = 0.343$, giving $1/0.657 = 1.52$.

Caution: The leave-one-out slope change is not the same quantity as Part 2′s “move up 10 °F” slope change. Point 1 moved up gives +0.825; point 1 removed gives −0.308. In the real cricket data the most influential single point is point 3 (+0.327), just ahead of point 1. Students who compare the two columns may be confused. The first asks “what could this point do?”; the second asks “what is this point doing?”

Q5.5 key.

Dataset	Biggest miss	Biggest slope change	Plant
1	21.5 (+16.3)	21.5 (+1.93)	21.5
2	16.7 (−21.4)	21.5 (−0.96)	16.7
3	21.2 (+12.2)	21.2 (+1.25)	21.2

They disagree in Dataset 2. The miss found the plant. The biggest slope change belongs to the real far point. Left-out slope change is exactly (distance from mean x) × (miss) / $(n \cdot \mathrm{SD}_x^2)$ – for point 1 of the crickets, $3.347 \times (-3.734)/40.54 = -0.308$ – so it rewards distance from the center as much as being off the trend. It measures influence, not error. The near-center plant is badly off the trend but has too little leverage to move the slope.